

IML-Lab -1

03-08-2026

Note: Do not share code with anyone.

Practice Question 1:

For each choice of dimension $d \in \{2^0, 2^1, 2^2, \dots, 2^{10}\}$, sample 100 points from the unit cube, and record the following average distances between all pairs of points, as well as the standard deviation of the distances.

- i. Squared Euclidean or ℓ_2 distance = $\|\mathbf{x} - \mathbf{y}\|_2^2 = \sum_j (x_j - y_j)^2$
- ii. ℓ_1 distance = $\|\mathbf{x} - \mathbf{y}\|_1 = \sum_j |x_j - y_j|$

Plot both the average and standard deviation as a function of d .

(You may wish to use `np.mean` and `np.std` to compute the statistics and `matplotlib` for plotting. You may find `numpy.random.rand` helpful in sampling from the unit cube.)

Write your observations about distance and standard deviation.

Practice Question 2:

Objective

The objective of this experiment is to understand the basic workflow of machine learning, including data loading, exploration, preprocessing, and visualization.

Dataset

- Iris Dataset (`sklearn.datasets`)

Tasks

- i. Import the following libraries:
 - NumPy

- Pandas
 - Matplotlib
 - Scikit-learn
- ii. Load the Iris dataset.
 - iii. Convert the dataset into a Pandas DataFrame.
 - iv. Display the first five rows of the dataset.
 - v. Determine the following:
 - Number of samples
 - Number of features
 - vi. Check whether the dataset contains any missing values.
 - vii. Calculate the following statistical measures for each feature:
 - Mean
 - Median
 - Maximum value
 - Minimum value
 - viii. Create the following visualizations:
 - (a) Histogram
 - (b) Scatter plot
 - (c) Box plot
 - ix. Write three observations based on the obtained results.

Include the output figure in your solution PDF (`Lab1_writeup.pdf`) and submit using classroom.